

Optimised sensor network for transmitter localisation and radio environment mapping

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Abstract: Location and power estimation of the radio transmitters is an efficient method of providing the radio environment maps in terms of database size and computation time. In this study, a novel efficient approach for estimating the power and location of active transmitters in the predefined geo-location is presented and the excellent results of this method are compared with other methods. Then, the raising question about number of required receiver sensors in the network area is discussed and an analytical relation is derived for this purpose. To perform experiment, the authors used distributed receivers in a site and conveyed their collected information about the received power in the specified bandwidth to the fusion centre. The results of this study are verified by both numerical simulations and experimental tests.

1 Introduction

Recent regulations for TV band communications and new trends of frequency access through geo-location databases relax the regulatory restrictions on cognitive radio (CR) systems [1]. This allows users to use secondary spectrum sensing databases from other devices, which eliminates the local spectrum sensing. This permission opens the issue of radio environment mapping (REM), recently introduced by some research groups [2–4]. Due to the dynamic nature of the primary user (PU) transmitters and wireless transmission channel, these maps are time varying and must be regularly updated. The preparation of these dynamic and geo-location-based maps require a significant number of spectrum sensors that makes the phone and/or Internet service providers to be a good choice. Some companies such as Google and Spectrum Bridge completed the tests of the 2009 FCC database administrator Call. In preparing these maps, the companies rely jointly on their large number of consumers who can act as sensors to measure the received signal power of the selected band in their geographical area and send their readings to the data fusion centres. In fusion centres, the collected measured values are used for estimating the transmitter power, location and the vacancy probabilities of certain bands in certain places [5].

In REM building the metrics of the white space spectrum detection must be satisfied [6]. Metrics to capture the sensing performance are the probability of detection, the probability of miss detection and the probability of false alarm. The situation in which the REM states that a PU is using a certain frequency band in a certain time range, while the spectrum in that frequency band is idle, is called a false alarm. On the other hand, the situation in which REM states that a frequency band is available, while it is in fact assigned, is called miss detection [7]. Miss detection causes interference with PU. The information rate in a specified bandwidth is an important parameter in communication systems, which is defined as the spectral efficiency. The spectral efficiency is affected by false alarms and miss detection. For an optimum detection performance, the probability of detection should be maximised while the probability of miss detection and false alarm is minimised [8]. In the draft of IEEE 802.22 standard [9], PUs should be detected within two seconds, and the probability of false alarm and miss detection should be less than or equal to 0.1 [10].

The performance of a local detector degrades in the presence of propagation impairments such as multipath fading and shadowing

[11–13]. This channel conditions can also lead in the hidden nodes, where a secondary transceiver is outside the hearing range of the primary transmitter but close enough to the primary receiver to provide interference. These problems can be relaxed with cooperative spectrum sensing (CSS) where geographically dispersed secondary users (SUs) cooperate in detecting a PU transmission through the exchange of measurement information among them before making a final decision. CSS can achieve different levels of performance through the exchange of different dataset. With hard decision fusion, every CR makes a local decision on the presence of primary user, then sends the binary decision to fusion centre (FC) for decision fusion. The voting rule is one of the easiest suboptimal merger rules [14]; it counts the number of nodes to agree on the presence of the signal, and compares the vote on a given threshold. In soft decision fusion, nodes send the summary statistics (locally measured energy) to the fusion centre. Then, the FC makes a global decision by linearly combining the local energy levels [15].

Similar to CSS, the REM building requires cooperation in sensor network. The PUs location information are the most important attributes of the REM. Considerable research has been done on localisation systems. Different categories of wireless localisation techniques have been introduced for various communication systems. Some techniques employed for wireless geo-location include the received signal strength (RSS) based systems [16], angle-of-arrival based systems [17], and propagation time based systems [18]. Due to the unavailability of any extra information about PU signals, the only localisation method that can be used in REM is RSS. This method has a precision about 3–30 m [15], that is not adequate especially for wireless local area network and ad-hoc CR networks. Moreover, this method requires at least four receiver node measurements and it is shown in this paper that cooperating with higher number of receiver nodes increases the precision. On the other hand, the growth of the number of cooperating sensor nodes in CSS increases the computational overhead and memory requirements.

In this paper, a novel cooperative localisation algorithm is presented. The experimental test data based on this algorithm shows that the localisation error regarding density of sensor placement is lower than one metre. Moreover, the effect of the number of cooperating sensors on localisation error and also on the CSS metrics is analysed and a practical equation for various

PU powers is obtained. For this purpose, in the following sections of this article, an overview of geo-location database and REM is given in Section 2. The environmental radio frequency modelling and target localisation method are analysed in Section 3. In Section 4, this analysis is extended to multi-target (multi-transmitter) cases and the simulation results are verified by experimental tests. In Section 5, the CSS and receiver operating characteristics (ROC) are reviewed and effect of cooperative nodes in sensing metrics is investigated. Finally, in Section 6, the required size of CR network is obtained. This issue is incorporated into REM construction and required number of sensors for providing accurate maps is determined.

2 Geo-location database and radio environment map

Virtual spectrum sensing and accessibility of database of white spaces, in the time and place, has an important role on development of CR technology; this requires developments of integrated database (REM) combined to environmental information, past experience and radio knowledge. The main objective is to enable CR to be situation-aware by simply referencing the REM, reducing the processing power and adaptation time of CR and producing low-cost large-scale CRs [19].

As shown in Fig. 1, a typical CR base-station includes a spectrum sensing module, REM, cognitive engine (CE), and a radio transceiver. The CE part requires the REM to be situation aware and to provide the utility function that best fits its current situation [20]. Moreover, in order to minimise harmful interference and maximise spectrum effectiveness, an optimisation approach can improve the performance of the CE.

2.1 REM data model

The current REM data model is composed of three main parts [21]:

- (i) Location and configuration information of transmitters, receivers, and possible dedicated spectrum sensors. This information can either be pre-configured or inferred.
- (ii) Information about the environmental features, such as statistical propagation methods.
- (iii) Radio interference fields (RIFs), which include spatial maps of RSSs, and other measurable quantities defined over a given area.

It should be noted that these information types are dependent to each other. For example, given transmitter locations and configurations together with a propagation model, the resulting RIF can be estimated. On the other hand, if there is no information on the transmitters, the measurement modules can obtain information about the RIF through power measurements on a specific channel, and these measurements can be used to infer the transmitter configurations in the region.

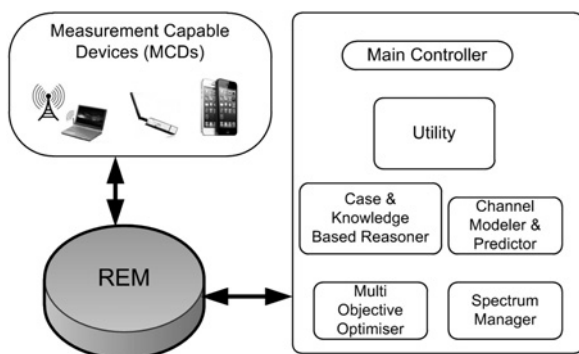


Fig. 1 Architectural diagram of REM-based CE

2.2 REM construction method

REMs have been introduced as integrated databases to provide an abstraction of the radio environment conditions [22–24]. Nevertheless, we focus only on deriving interference level at each point of the CRN coverage region. This is also interpreted as interference cartography (IC). IC forms a map of signal strength of RF as a function of spatial coordinates in a predefined area.

In the literature, REM construction methods can be divided into two classes: spatial statistics based methods [25–27], and transmitter location determination based approaches [28], which are also referred to as direct and indirect methods, respectively.

Spatial statistics gives information on the statistical properties of a given region utilising the spatial correlational structure of this area. Interpolation is based on the basic principle that geographically closer locations are compared with more distant places. Common interpolation techniques for IC are Kriging method, inverse distance weighted interpolation, and nearest neighbour interpolation. The frequency or volume of measurements, required accuracy in measurements, and density of measurement points are key factors for determining the performance of each technique.

Transmitter location determination based methods use the availability of the active transmitter information to construct REM. These methods first focus on localising the transmitter(s) and deriving their properties. Then, they estimate the signal strength at each point by applying the propagation modelling. In [25], authors study the effect of transmitter properties on the signal strength in a CRN using the second-order statistics. In [28], an image processing based technique is applied, which identifies the transmitters in the system and estimates their parameters based on RSS measurements from sensors. We follow this method and first focus on estimating the active transmitter location and then on constructing the REM.

3 Environment model and target localisation

In this section, we characterise the variation in received signal power over distance due to path loss and shadowing. Path loss is produced by dissipation of the power radiated by the transmitter as well as the effects of the propagation channel. Path loss models commonly suppose that path loss is the same for a given transmit-receive distance. On the other hand, shadowing is produced by obstacles between the transmitter and receiver that absorb or reflect energy. Variation due to path loss takes place over very large distances, whereas variation due to shadowing happens over distances proportional to the dimensions of the obstructing object. Since variations due to path loss occurs over relatively large distances, this variation is sometimes referred to large-scale propagation effect. There is another variation due to the constructive and destructive addition of multi-path signal components which occurs over very short distances and are referred to as small-scale propagation effects or multi-path fading [29].

3.1 Path loss model

Due to the complexity of signal propagation it is hard to achieve a single model for featuring path loss precisely across a range of various environments. However, for general trade off analysis of different system designs, it is often best to apply a simple approximate model that captures the essence of signal propagation without resorting to complex path loss models. Thus, the following simple model for path loss as a function of distance is utilised for system design

$$P_r = KP_t \left(\frac{d_0}{d} \right)^\alpha \quad (1)$$

Hence, the dB attenuation is

$$P_r(\text{dBm}) = P_t(\text{dBm}) + K(\text{dB}) - 10\alpha \log_{10}\left(\frac{d}{d_0}\right). \quad (2)$$

In this approximation, P_r is the received power at the receiver, P_t presents the transmitter's output power, K presents a unitless constant that depends on the antenna characteristics and the average channel attenuation, d_0 is a reference distance for the antenna far-field, and α presents the path loss exponent.

3.2 Shadow fading

In addition to path loss, a signal typically experiences random variation due to blockage from objects in the signal path, yielding a random variation about the path loss at a known distance. Moreover, changes in reflecting surfaces and scattering objects can also cause random variation in the path loss. Therefore, a model for random attenuation representing the influence of these effects is also required. Since the location, size, and dielectric properties of the blocking objects as well as the changes in reflecting surfaces and scattering objects are usually unknown, statistical models should be used to characterise such attenuations. The most common model for this extra attenuation is log-normal shadowing.

$$P_r(\text{dBm}) = P_t(\text{dBm}) + K(\text{dB}) - 10\alpha \log_{10}\left(\frac{d}{d_0}\right) + \psi_{\text{dB}} \quad (3)$$

where ψ_{dB} is a Gaussian-distributed random variable with zero mean and $\sigma_{\psi_{\text{dB}}}^2$ presents the variance.

3.3 Multi-path fading

If a single pulse is transmitted over a multi-path channel, the received signal reveals as a pulse train with each pulse in the train corresponding to the line of sight (LOS) component or a distinct multi-path component related to a distinct scatterer or cluster of scatterers. The effects of multi-path consists constructive and destructive interference, and phase shifting of the signal. In the urban areas where there are many scatterer, it is observed that the magnitudes of the arriving signals by the various paths have a distribution known as the Rayleigh distribution. This type of

fading is known as Rayleigh fading. While, when one component (often, but not necessarily, a LOS component) dominates, a Rician distribution provides a more accurate model; and this is referred as Rician fading. If there is multiple independent transmitters in a channel, then the sum of multiple independent and identically distributed Rayleigh-fading signals have Nakagami distributed signal amplitude. Under Rayleigh fading, signal to noise ratio (γ) would have an exponential distribution [30] as follows

$$f(\gamma) = \frac{1}{\bar{\gamma}} \exp\left(-\frac{\gamma}{\bar{\gamma}}\right) \quad \gamma \geq 0. \quad (4)$$

In this case, $E[\gamma] = \bar{\gamma}$, $\text{Var}[\gamma] = \bar{\gamma}^2$. This type of fading has a dominant effect on mobile wireless communication system, where the number of samples in any point is limited. In our case, where the receiver nodes are fixed and we do not have any restriction on the number of sample, the dominant factor is shadowing. Accordingly, in the following sections we focus on shadowing.

3.4 Target localisation

After an overview of preliminary subjects, now we can describe the transmitter localisation method. Assume that in a predefined area, some receiver sensors are placed in a random or mesh-type manner. For the first step a transmitter is placed in an interior point of this area, and the sensors send their received RF power measurements in a fixed bandwidth to the FC. The aim of the FC is to estimate the transmitter's location and power. The proposed system schematic is shown in Fig. 2.

By using (2), we can write the following relation

$$P_r(\text{dBm}) = P_t(\text{dBm}) + K(\text{dB}) - 10\alpha \log_{10}\left(\frac{\sqrt{(x_i - x_t)^2 + (y_i - y_t)^2}}{d_0}\right). \quad (5)$$

where, (x_i, y_i) and (x_t, y_t) are the spatial coordinates for the i^{th} receiver and transmitter, respectively. For an arbitrary reference distance d_0 , the received power is

$$P_{r_0}(\text{dBm}) = P_t(\text{dBm}) + K(\text{dB}). \quad (6)$$

We can rewrite (5) by using the path loss function ($\text{PL}_i = P_{r_i} - P_{r_0}$) as follows

$$x_t^2 + y_t^2 - 2x_i y_t + x_i^2 + y_i^2 = d_0^2 10^{\text{PL}_i/5\alpha}. \quad (7)$$

After some manipulations and considering four receiver data (measured power and its coordinates), the following linear matrix equations are obtained

$$\begin{bmatrix} 2(x_2 - x_1) & 2(y_2 - y_1) & 10^{-P_{r_2}/5\alpha} - 10^{-P_{r_1}/5\alpha} \\ 2(x_3 - x_1) & 2(y_3 - y_1) & 10^{-P_{r_3}/5\alpha} - 10^{-P_{r_1}/5\alpha} \\ 2(x_4 - x_1) & 2(y_4 - y_1) & 10^{-P_{r_4}/5\alpha} - 10^{-P_{r_1}/5\alpha} \end{bmatrix} \begin{bmatrix} x_t \\ y_t \\ d_0^2 10^{\text{PL}_0/5\alpha} \end{bmatrix} = \begin{bmatrix} x_2^2 + y_2^2 - x_1^2 - y_1^2 \\ x_3^2 + y_3^2 - x_1^2 - y_1^2 \\ x_4^2 + y_4^2 - x_1^2 - y_1^2 \end{bmatrix}. \quad (8)$$

Equation (8) is a linear matrix equation as ($\mathbf{A}\boldsymbol{\theta} = \mathbf{Y}$), where $\mathbf{A}$ and $\mathbf{Y}$ are determinant (system parameters and observation) vector and matrix, respectively, and $\boldsymbol{\theta}$ is the equation variable vector (transmitter characteristics x_t, y_t, PL_0) that must be found. Shadowing and measurement tolerances cause some error in this equation. This is a linear regression equation, and a minimised mean square error (MSE) solution needs some extra measurements. There are various solving methods such as adaptive

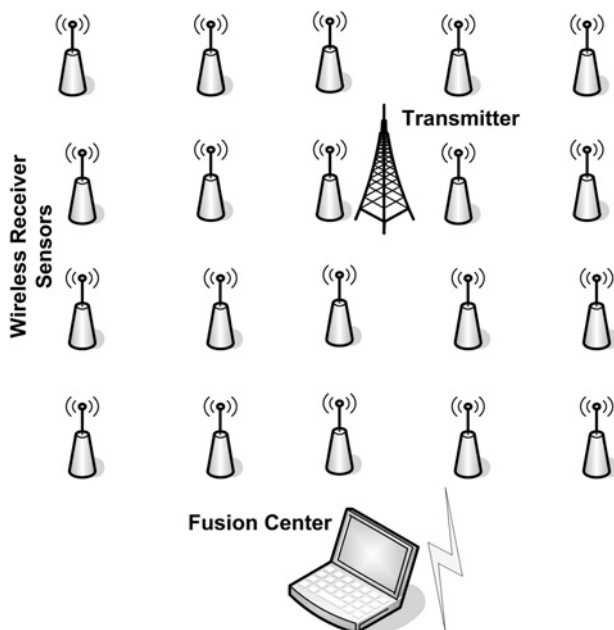


Fig. 2 Installed transmitter localisation system

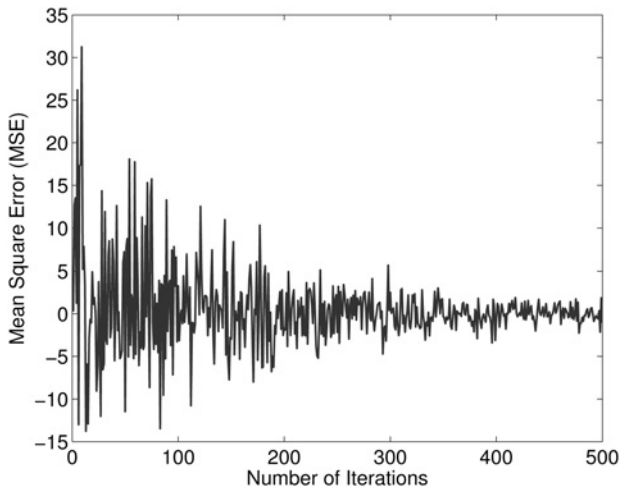


Fig. 3 MSE of transmitter localisation by iterative adaptive method

iterative method or batch method [31]. In adaptive iterative method for solving linear equation $\mathbf{Y} = \mathbf{A}\boldsymbol{\theta} + \mathbf{E}$, an initial guess is given for $\boldsymbol{\theta}$, then the value of $\boldsymbol{\theta}$ in the subsequent steps is changed in a manner that the MSE is decreased by steepest decent. The resulting updating equation is

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i + \mu \mathbf{A}^T \mathbf{e}_i. \quad (9)$$

where $\mathbf{e}_i = \mathbf{Y}_i - \mathbf{A}\boldsymbol{\theta}_i$ is the instantaneous error value and μ is the step size. For testing this method, we first consider a simulation setup. In a predefined area with a transmitter located in an interior point of this area, signal's strengths are measured in some random selected points, and by using iterative algorithm of (9) the location and the power of the transmitter are estimated. Consider that, in each step, the data of four receiver are used for updating equation. The result of simulation with additive white Gaussian noise (AWGN) is shown in Fig. 3.

This localisation method is also verified by realistic experiments. For this purpose, a mesh type receiver network is provided. The receivers are placed in a 7×7 node network with 5 m mesh side size and a transmitter is located at a random point of this area. NI-USRP devices are used as receivers and a wireless microphone transmitter in 800 Mhz free band is used as the transmitter. The setup and the devices are shown in Fig. 4.

In USRP receivers, the measured spectrum power in 800 Mhz band with 200 KHz bandwidth is averaged and transferred to the FC. First, the gathered data is used to estimate the path loss function order. The results of this experiment and the fitted function by optimised order ($\alpha = 3.45$) is shown in Fig. 5. Then in FC, the gathered data from all receivers are analysed by iterative algorithm and the transmitter characteristics are estimated. The results of localisation algorithm is shown in Fig. 6. In Fig. 6a, the algorithm status in every step with exact transmitter location

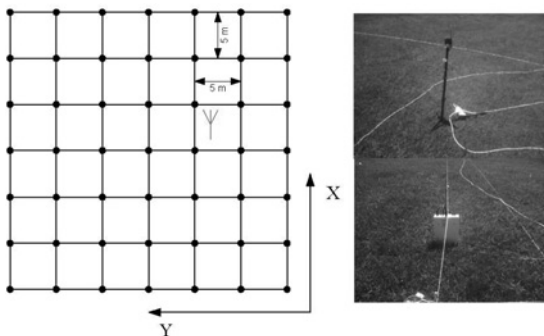


Fig. 4 Setup for transmitter localisation

(x_i, y_i) is shown, and in Fig. 6b localisation tolerances after the convergence of the algorithm are shown.

This localisation algorithm has a major drawback: it requires the collection of all sensor data before starting the estimation process. This drawback can be tackled by an alternative method. As shown in (8), any guess for transmitter localisation requires four sensor data. Suppose that the network has a common random access data transfer channel to FC, and in each time slot only one sensor can transfer its measurements. Thus in the first three time slots no estimation can be computed, and after fourth slot the estimation process is started. In time slot n , we have $\binom{n}{4}$ combinations of sensor data that can be considered for (8).

$$\begin{cases} A_1 \boldsymbol{\theta} + e_1 = Y_1 \\ \vdots \\ A_n \boldsymbol{\theta} + e_n = Y_n \end{cases} \quad (10)$$

The MSE is defined as follows

$$\text{MSE} = (A_1 \boldsymbol{\theta} - Y_1)^2 + \dots + (A_n \boldsymbol{\theta} - Y_n)^2 = \sum_{i=1}^n (A_i \boldsymbol{\theta} - Y_i)^2. \quad (11)$$

Thus, $\boldsymbol{\theta}$ that minimise this error can be obtained by using the first order derivative

$$\frac{\delta \text{MSE}}{\delta \boldsymbol{\theta}} = \sum_{i=1}^n (2A_i^T (A_i \boldsymbol{\theta} - Y_i)) = 0 \implies \boldsymbol{\theta}_{\text{opt}} = \frac{\sum A_i^T Y_i}{\sum A_i^T A_i}. \quad (12)$$

We observe that by increasing the amount of received data, the number of combinations, and as a result, the computation complexity increases significantly. This increase can be controlled by selecting finite number of data at each step. The result of this method is shown in Fig. 7. We should mention that because of the random access channel, the selection of nodes and thus the SNR of selected nodes are random, which leads to differing results for the same setup. Thus, we give the average of 25 experiments. This simulation (by using experiment data) shows that the data of approximately 20 sensors is sufficient for transmitter localisation.

In the final step of this section, a comparison between the proposed adaptive method and well known Kalman filtering method is presented. The Kalman filtering method is a classic recursive tracking method that have a good accuracy in problems with AWGN [32]. In REM, it is considered that the smart phones of the customers act as the spectrum sensors. Thus, due to the recursive manner of the Kalman filtering, we cannot use this

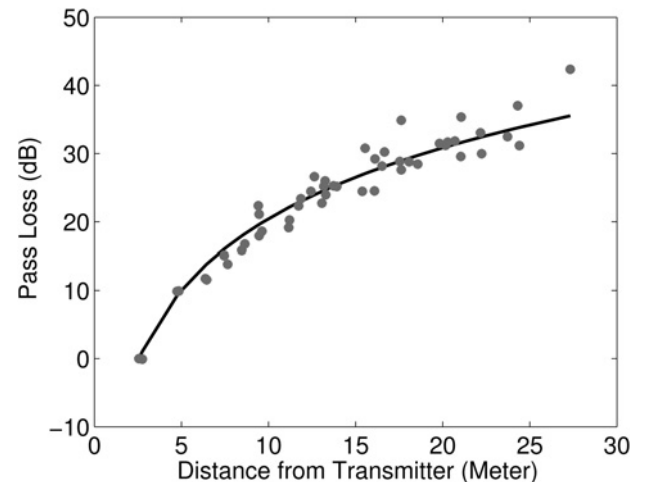


Fig. 5 Path loss function estimation

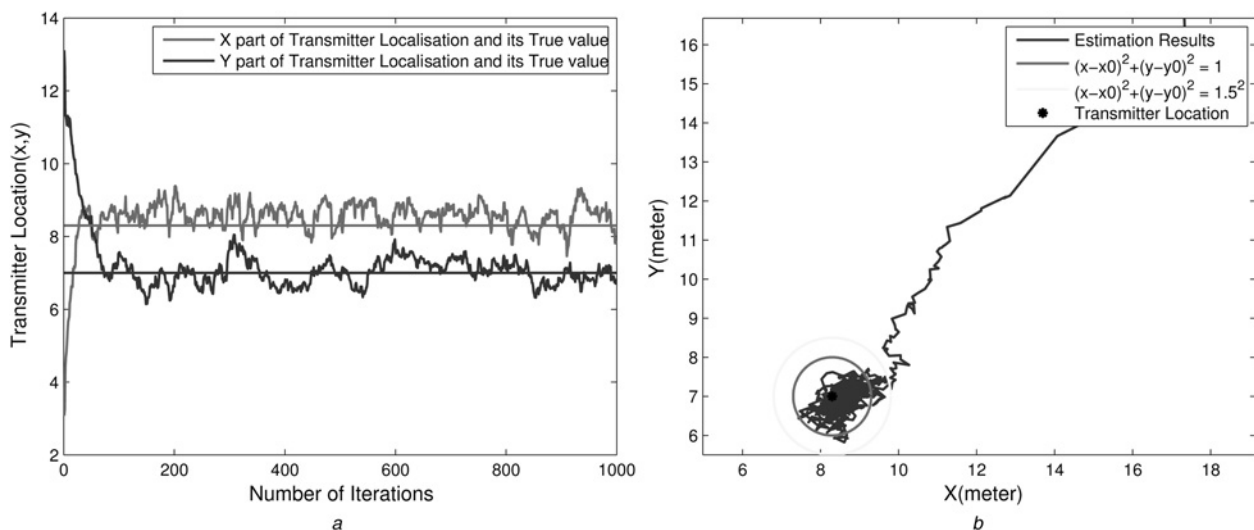


Fig. 6 Results of localisation algorithm

a Localisation algorithm convergence
b Localisation algorithm tolerances

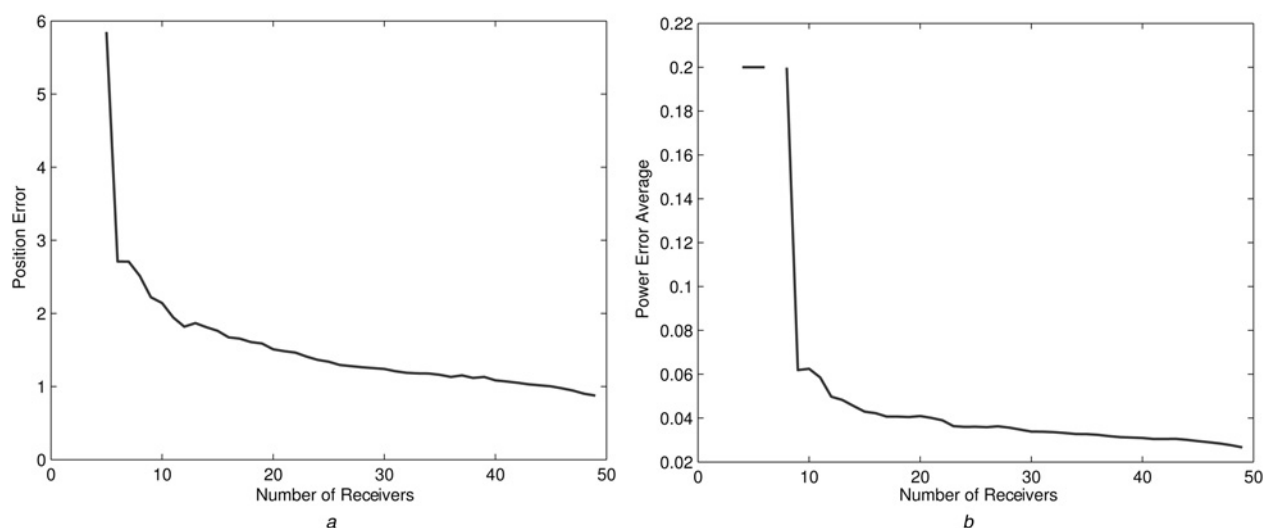


Fig. 7 Localisation algorithm accuracy against number of cooperating receiver

a Transmitter localisation error
b Power error

method for transmitter localisation. Moreover the multiplicative measurement noise of the RSS based localisation methods that is shown in (8) declines the performance of the Kalman filtering. However, in the implementation of Kalman filtering, (8) is used as observation equation and the following equation

$$\begin{bmatrix} x_t(k+1) & y_t(k+1) & 10^{PL(k+1)} \end{bmatrix}' = \begin{bmatrix} x_t(k) & y_t(k) & 10^{PL(k)} \end{bmatrix}' \quad (13)$$

in which k is time step, is used as state equation. Fig. 8 shows comparison results of the proposed method and Kalman filtering method. This figure shows that our proposed method outperform Kalman filtering.

4 Multi-target localisation

If we have more than one active transmitter (in a specific frequency band), we need to employ a different localisation method. In this case, the average spectral power density (SPD) function of

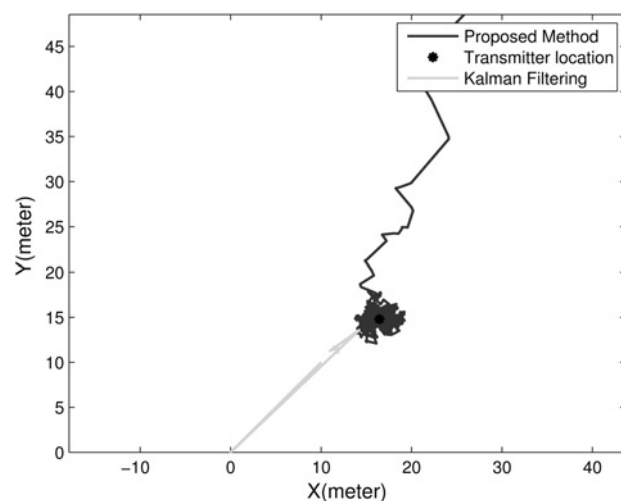


Fig. 8 Results of Kalman filtering and proposed adaptive localisation methods

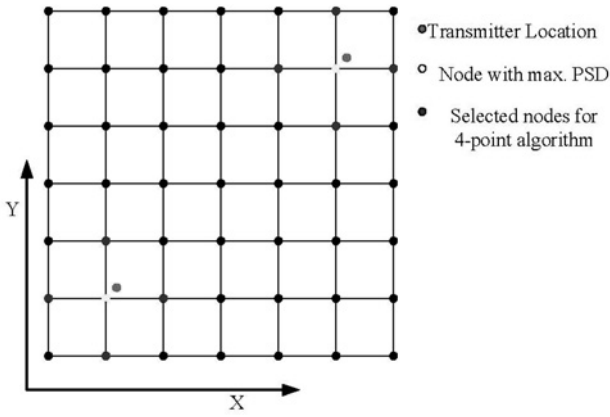


Fig. 9 Schematic of implemented sensor network for multi-target localisation

measurement region has more than one maximum point. Thus the localisation of transmitters is performed in two steps. In the first step, the sensors whose measurements are higher than their neighbours are selected. In the second step, the data of these sensors (maximum points) and their neighbours are used for estimation. The number of selected sensors are limited and the localisation algorithm of (12) is preferred.

To verify the performance of this method, a receiver network setup (mesh placement) with 7×7 sensors is prepared. In this setup, two transmitters are placed within this area. Transmitters are implemented by wireless microphone in 700 MHz public safety group band (in mute mode) and a NI-USRP device in continuous transmission mode in 700 MHz band. These conditions ensure constant power operation for transmitters. Receivers are implemented by NI-USRP devices in receiving mode, and the required control connections are prepared via local area network facilities of these devices. After gathering measurements of 49 sensors in FC, the maximum of measured powers in each row and column of mesh is found. The overlap points of row and column maximums give the maximum points of the SPD function of the region. The schematic of the network with highlighted transmitter location and local maximum points and their neighbours is shown in Fig. 9. The resulting transmitter location and REM of region is also appeared in Fig. 10. Due to transmitters' power influence on the measured power of other transmitter neighbours, the localisation error is higher than the one-transmitter case. This effect can be neglected in far distance transmitters.

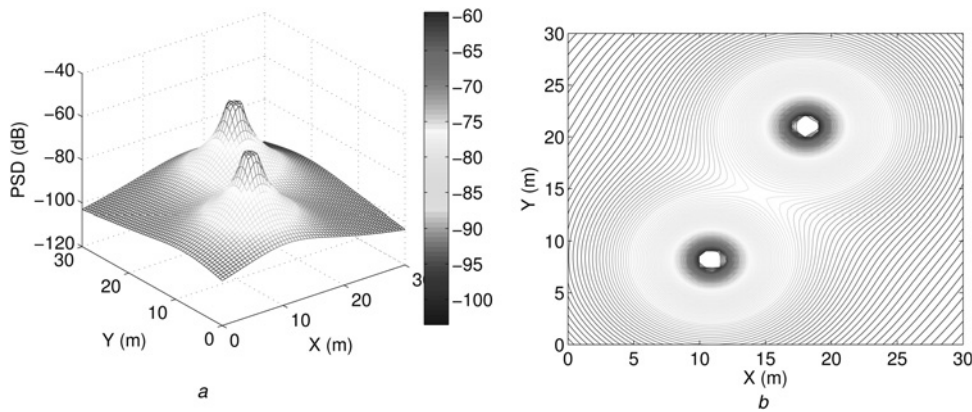


Fig. 10 Radio environment map of network area

a 3-dimensional view
b Equal power contours

5 CSS and detection performance metrics

5.1 Overview

The goal of spectrum sensing is to decide between two binary hypotheses. If H_0 represents the absence of primary signal and H_1 represents the presence of primary signal, then the received signal in a secondary node at both hypotheses is

$$\begin{cases} H_0: x(n) = v(n) \\ H_1: x(n) = hs(n) + v(n) \end{cases} \quad n = 1, 2, \dots, M \quad (14)$$

where $s(n)$ is the primary user's transmitted signal, $v(n)$ is AWGN, and h is the amplitude gain of the channel. M is the number of samples that must satisfy the Shannon limit ($f_s = (1/T_s) \geq 2WM = (T/T_s) \geq 2TW$, where f_s is sampling rate, T is sensing time and W is bandwidth of signal). The decision rule in energy detection approach is given by

$$T(\mathbf{x}) \triangleq \sum_{n=1}^M |x(n)|^2 \underset{H_0}{\overset{H_1}{\geq}} \lambda \quad (15)$$

where $\mathbf{x} = [x(1) \ x(2) \ \dots \ x(M)]^T$, $T(\mathbf{x})$ is the test statistics, and λ is the corresponding test threshold. Although $T(\mathbf{x})$ has a chi-square distribution [18], according to the central limit theorem $T(\mathbf{x})$ is asymptotically normally distributed if M is large enough ($M \geq 20 - 30$ is sufficient in practice) [33]. Specifically, for large M , we can model the statistics of $T(\mathbf{x})$ as follows

$$T(\mathbf{x}) \sim \begin{cases} \mathcal{N}(M\delta_v^2, 2M\delta_v^2) & \text{under } \mathcal{H}_0 \\ \mathcal{N}(M\delta_v^2 + Mp_s, 2M\delta_v^2 + 4M\delta_v^2 p_s) & \text{under } \mathcal{H}_1 \end{cases} \quad (16)$$

where $p_s = \|s\|^2/M$ represents the average power of the primary signal. In this way, for large M , the probability of false alarm, P_f , and the probability of detection, P_d , can be approximated as

$$P_f = P(\mathcal{H}_1|\mathcal{H}_0) = Q\left(\frac{\lambda - M\delta_v^2}{\delta_v\sqrt{2M}}\right) \quad (17)$$

$$P_d = P(\mathcal{H}_1|\mathcal{H}_1) = Q\left(\frac{\lambda - M\delta_v^2 - Mp_s}{\delta_v\sqrt{2M\delta_v^2 + 4Mp_s}}\right) \quad (18)$$

where $Q(x)$ is the tail probability of a zero mean unit variance Gaussian random variable. Now, for a predefined probability of false alarm, the probability of detection for receiver sensors as a

function of SNR ($\text{SNR} = p_s/\delta_v^2 = \|s\|^2/M\delta_v^2$) can be computed as

$$P_d = Q\left(\frac{Q^{-1}(P_f) - \sqrt{M/2} \text{SNR}}{\sqrt{1 + 2\text{SNR}}}\right). \quad (19)$$

These equations can be extended to soft decision CSS where, for overcoming the shadowing and multi-path fading problems, the local energy levels of some CR sensors are linearly combined in a FC. In this system, the test statistic is, $L(y) = w^T y \underset{H_0}{\underset{H_1}{\geq}}$ λ , where $w = [w_1, w_2, \dots, w_N]$ is the weight vector of cooperating sensors in fusion and $y = [y_1, y_2, \dots, y_N]$ is their local energies. Since the linear combination of several Gaussian random variables is still Gaussian, the performance of the linear detector can be evaluated as

$$P_f = P(\mathcal{H}_1|\mathcal{H}_0) = Q\left(\frac{\lambda - \mu_0^T w}{\sqrt{w^T \Sigma_0 w}}\right) \quad (20)$$

$$P_d = P(\mathcal{H}_1|\mathcal{H}_1) = Q\left(\frac{\lambda - \mu_1^T w}{\sqrt{w^T \Sigma_1 w}}\right) \quad (21)$$

where, $\mu_0 = [\mu_{10}, \dots, \mu_{M0}]^T$, $\mu_1 = [\mu_{11}, \dots, \mu_{M1}]^T$ where $\mu_{i0} = M\delta_{vi}^2$, $\mu_{i1} = M\delta_{vi}^2 + Mp_{si}$ are the mean values of i th sensor's output, and

$$\Sigma_0 = \begin{bmatrix} \delta_{i0}^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \delta_{N0}^2 \end{bmatrix}, \quad (22)$$

$$\Sigma_1 = \begin{bmatrix} \delta_{i1}^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \delta_{N1}^2 \end{bmatrix} \quad (23)$$

where $\delta_{i0}^2 = 2M\delta_{vi}^4$, $\delta_{i1}^2 = 2M\delta_{vi}^4 + 4M\delta_{vi}^2 p_{si}$ are the variances of i th sensor outputs under $\mathcal{H}_0$ and $\mathcal{H}_1$ hypothesis.

5.2 Effect of cooperative sensor number

The number of cooperating sensors is an important parameter in CSS and REM since the computational complexity, database size and energy consumption depends on it. Unlike most of the mentioned characteristics, the performance of the spectrum sensing is improved by increasing the number of sensor number. This point is shown in Figs. 11 and 12 for a sample network that is defined

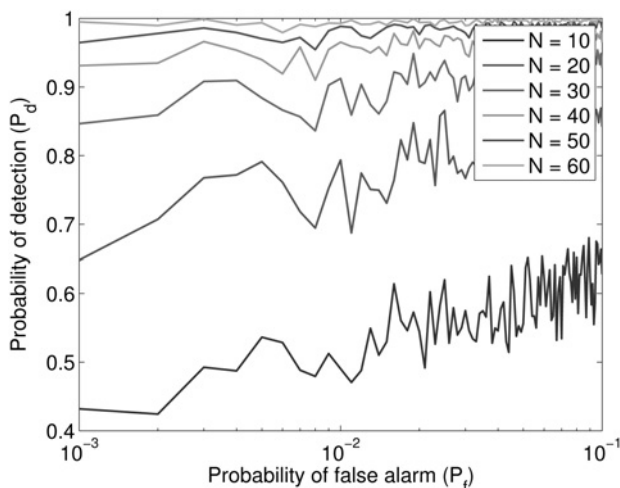


Fig. 11 ROC curves for various number of cooperating sensors

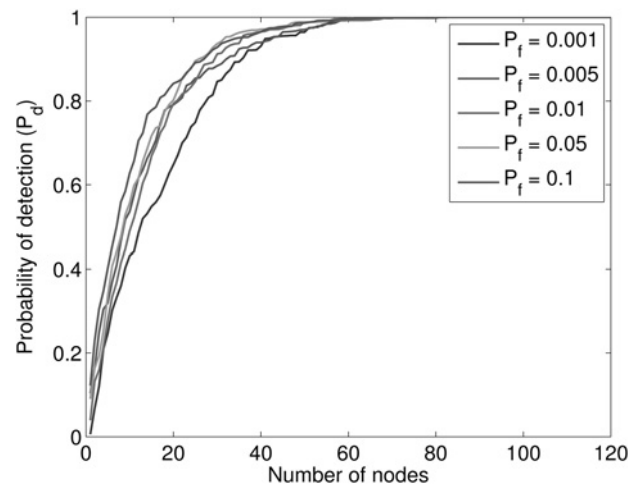


Fig. 12 Probability of detection against number of cooperating sensors

in the next paragraph. On the other hand, PU transmitters have different power ranges, from 1–10 watt WiFi transmitters up to kilowatt TV base stations. It is obvious that the network area and the number of distributed cooperating receiver sensors in these cases, for providing REM, are different. Thus, for generality, we change the issue of total number of sensors, to the density of sensors per estimated power of transmitter. In the next section this method is described.

In the soft decision CSS, the effect of low SNR sensors in decision is negligible. Thus, we can define a circle around primary transmitter such that at interior points of it, the SNR(dB) be greater than some predefined value (in this research, 0 dB) and in cooperation, the sensors that located in this area is considered. For analysing the effect of random placed sensor number on sensing metrics a simulation setup is made. The transmission power of the wireless transmitter is assumed to be -40 dB while the noise floor of the medium is taken as -100 dB. In the circle around the transmitter with SNR higher than 0 dB, the receiver sensors are placed in a random manner. The shadowing effect, with log-normal distribution is also considered. Then, the receiver operating characteristics (ROC) curves against the number of cooperating nodes are obtained and shown in Fig. 11. In this experiment, like the previous ones, because of the random placement of sensors, the experiments are repeated 25 times and the average of results is shown. Another version of Fig. 11 is shown in Fig. 12, where the probability of detection is plotted against the number of cooperating nodes for different values of false alarm probabilities.

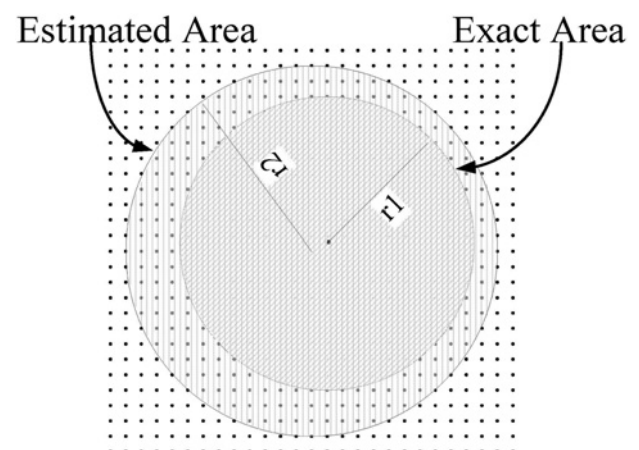


Fig. 13 Correct detection area and estimated detection area

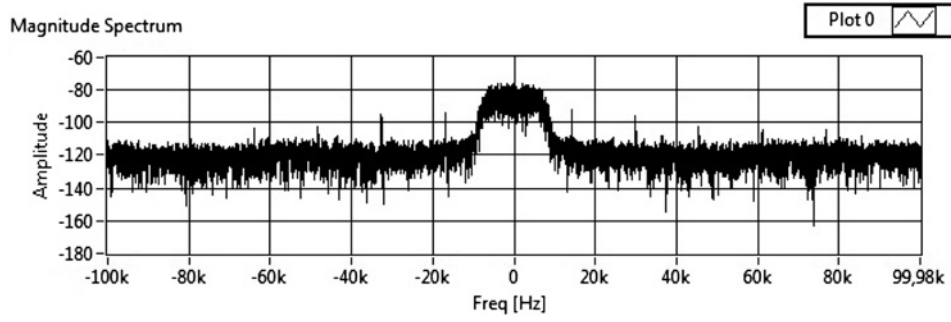


Fig. 14 Magnitude spectrum of a receiver sensor

6 Accurate network size

In Sections 3 and 4, the primary transmitter location and power in single and multi-target states are estimated and the number of required sensor to receive a predefined error level is obtained. In Section 5, the detection probability in spectrum sensing process and also the relationship between the density of cooperative receiver sensors and detection probabilities are obtained.

In this section, the aim is to extend these issues to general case. Consider that the number of cooperative CR sensors for detection of primary transmitters with SNR higher than a specified value SNR_{th} is required. As the first step, the metrics for spectrum sensing must be defined. For example, in the IEEE 802.22 standard, these metrics are $P_d \geq 0.9$ for $P_f = 0.1$. Figs. 11 and 12 show that the minimum number of required sensors in sensing area is approximately 30 ($N_0 = 30$). The relationship for sensing area by using (1) is obtained as

$$S_s = \pi d^2 = \pi \times (d_0)^2 \times 10^{(SNR_0(dB) - SNR_{th}(dB))/5\alpha} \quad (24)$$

where SNR_0 is the signal to noise ratio in a predefined distance (d_0) from transmitter. Thus, the sensor density is

$$S_D(\text{Sensor density}) = \frac{N_0}{S_s} \quad \text{Sensor/m}^2. \quad (25)$$

This sensor density is based on the transmitter localisation results. On the other hand, the localisation accuracy (as shown in Fig. 7) is dependent on the number of receiver sensors. Therefore, restricting the number of sensors increases the localisation error, and this error creates some difference between correct and estimated detection areas as shown in Fig. 13.

Our measurements in Section 3 show that the transmission power estimation error is usually high compared with the localisation errors, hence the estimated area and correct area are overlapping circles. Thus, some of the cooperating sensors in localisation are out of the effective area and have negligible effects. Therefore, we can calculate the effective number of sensors from the following equation

$$N_{eff} = \frac{S_{overlap}}{S_{est.}} N_0. \quad (26)$$

where the ratio of overlapping area to the estimated area can be computed using the estimated power error (Fig. 7b) as follow

$$\frac{S_{overlap}}{S_{est.}} = \frac{\pi r_1^2}{\pi r_2^2} = \left(\frac{r_1}{r_2}\right)^2 = \sqrt{\frac{P_t(\text{correct})}{P_t(\text{est.})}} \approx \sqrt{1 - \epsilon} \quad (27)$$

where

$$\epsilon = P_t(\text{est.}) - P_t(\text{correct})$$

To verify these relations, we use the results of transmitted signal and noise power of an experiment as shown in Fig. 14.

The SNR differences of (24) because of this figure is approximately 20 dB. Fig. 7b shows that $\epsilon = 0.05$. Thus the required sensor density of this network is

$$S_D = \frac{30\sqrt{1-0.05}}{\pi \times 100} = 0.093.$$

This result approximately equals one sensor in 10 m, that meets our implemented network inter-sensor distances.

7 Conclusion

In this paper, the CSS and REM is analysed. In the proposed transmitter localisation method, the impact of environmental parameters such as shadowing and atmospheric noise are considered. The experimentally verified results show less than one metre tolerance that shows better performance than existing methods. Then some preliminary REM in single and multi-transmitter conditions are provided. The effect of the number of cooperating sensors in REM and detection probability are described in detail and an efficient frontier for network size is obtained ($N = 30$). In addition, a new approach for calculation the network size (receiver sensor density) is introduced. In future work, the tracking of moving transmitters in outdoor and indoor conditions for providing REM should be considered.

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